UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

ECE408 - CS483 - CSE408 Spring 2026
Applied Parallel Programming

Lecture 9: 2D Tiled Convolution Kernel; Reuse Analysis

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2/17/2026

Course Reminders

- Lab updates
  - Lab 4 is due this week on **Friday**
- Project milestone 1 will be released soon
- Take a note of Midterm 1 date/time
  - **When:** March 3rd, 7-10pm
  - **Where:** Your specific room assignment will be posted in Canvas
  - **What:** Lectures 1-11, Labs 1-4
  - **How:** paper-based
  - **Alternative Exam Time:** as arranged, email me by 2/24/26 if you have a valid conflict
  - **Study materials:** now posted on Canvas
  - **Exam review session:** 3:00-5:30PM, Saturday, February 28th, in 1015 ECEB

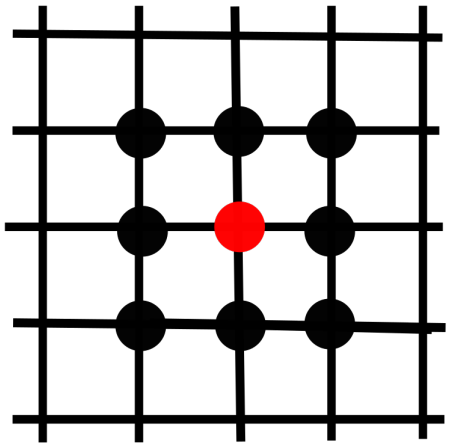
Today's Objectives

- To learn more about tiled convolution algorithms
  - Some intricate aspects of tiling algorithms
  - Output tiles versus input tiles
  - To prepare for Lab 4

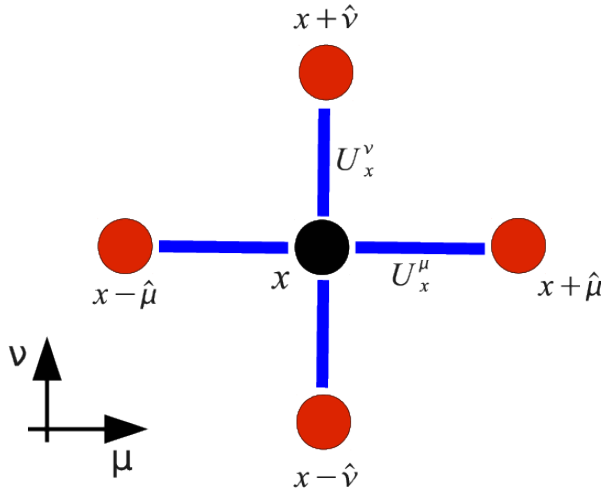
- To learn more about the analysis of tiled convolution/stencil algorithms

Stencil Algorithms

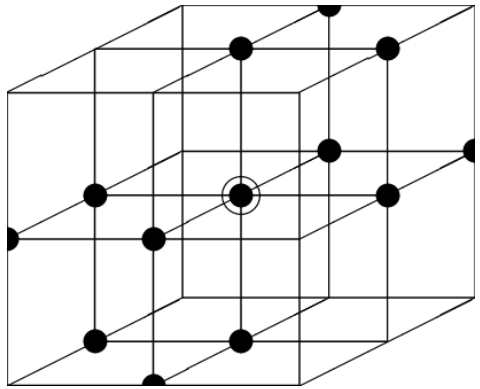
- Numerical data processing algorithms which update array elements according to some fixed pattern, called a *stencil*
  - Convolution is just one such example



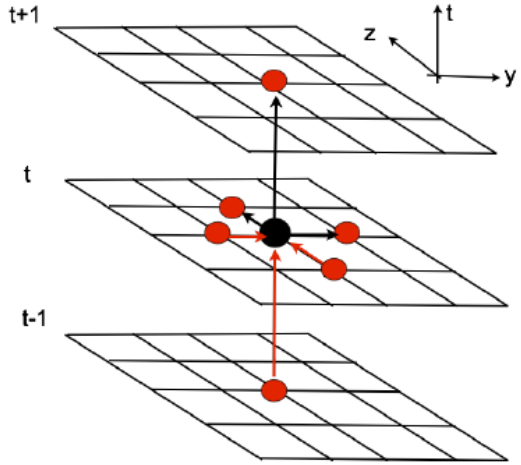
2D convolutional kernel (9-point 2D stencil)



Nearest neighbor lattice Dirac operator (5-point 2D stencil)

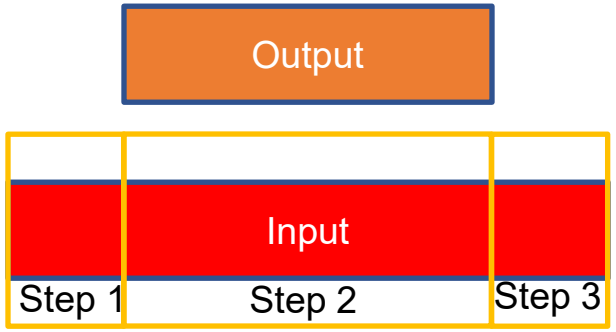


Finite difference stencil for 3D explicit time-marching (13-point 3D stencil)



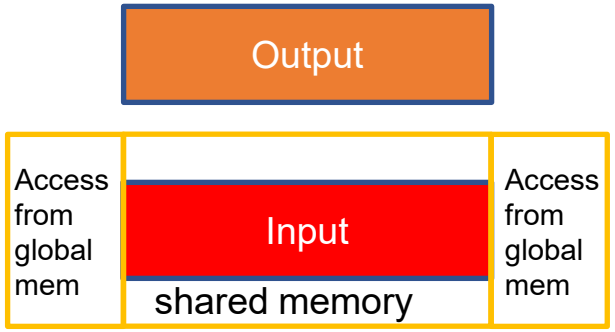
The Wilson-Dslash operator (4D stencil)

Review: Three Tiling Strategies



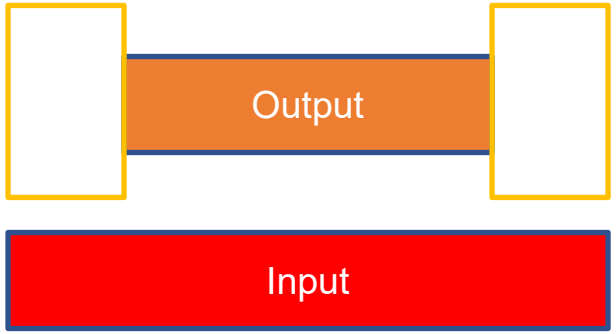
Strategy 1

1. Block size covers **output** tile
2. Use multiple steps to load input tile



Strategy 3

1. Block size covers **output** tile
2. Load only "core" of input tile
3. Access halo cells from global memory



Strategy 2

1. Block size covers **input** tile
2. Load input tile in one step
3. Turn off some threads when calculating output

Review: What Shall We Parallelize?

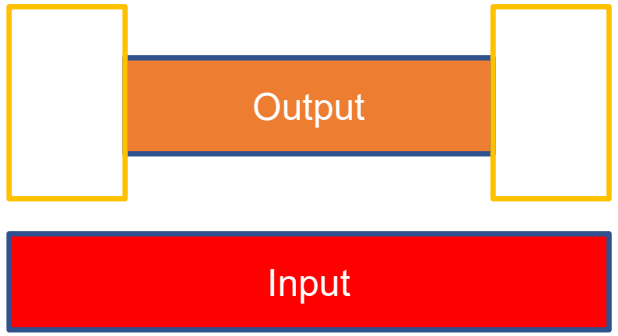
- In other words,
  - **What should one thread do?**

- **One answer:**
  - (same as with vector sum and matrix multiply)
  - **compute an output element!**
    - **Strategy 1 & 3**

- **Is that our only choice? (What about Strategy 2?)**

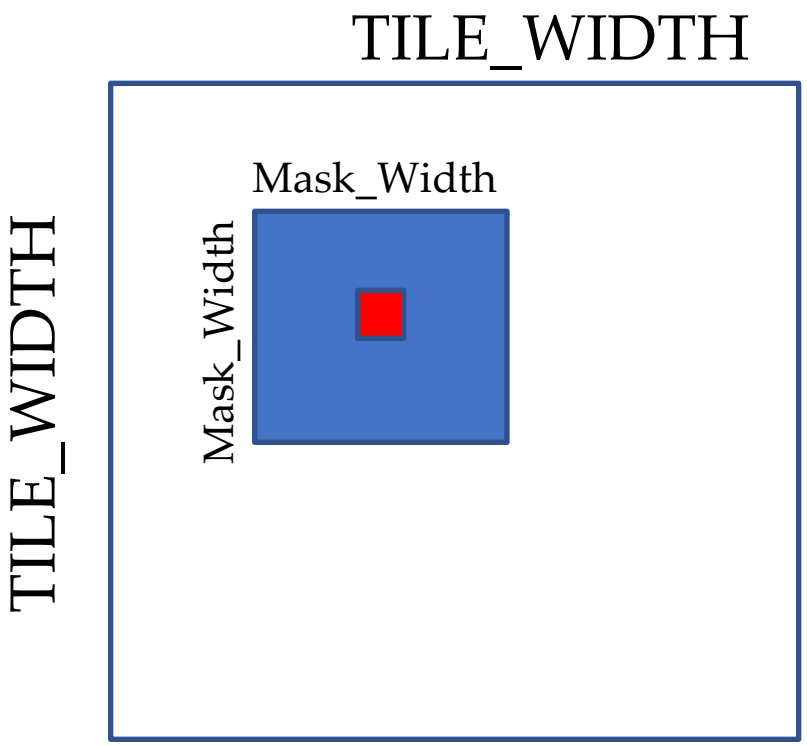
Strategy 2: Parallelize Loading of a Tile

- Alternately,
  - Thread block matches input tile size
  - Each thread loads one element of input tile
  - Some threads do not participate in calculating output
- Advantage:
  - **No** branch **divergence for load** (high latency).
  - **Avoid narrow global access** (2 × halo width).
- Disadvantage:
  - Branch **divergence for compute** (low latency).



Parallelizing Tile Loading

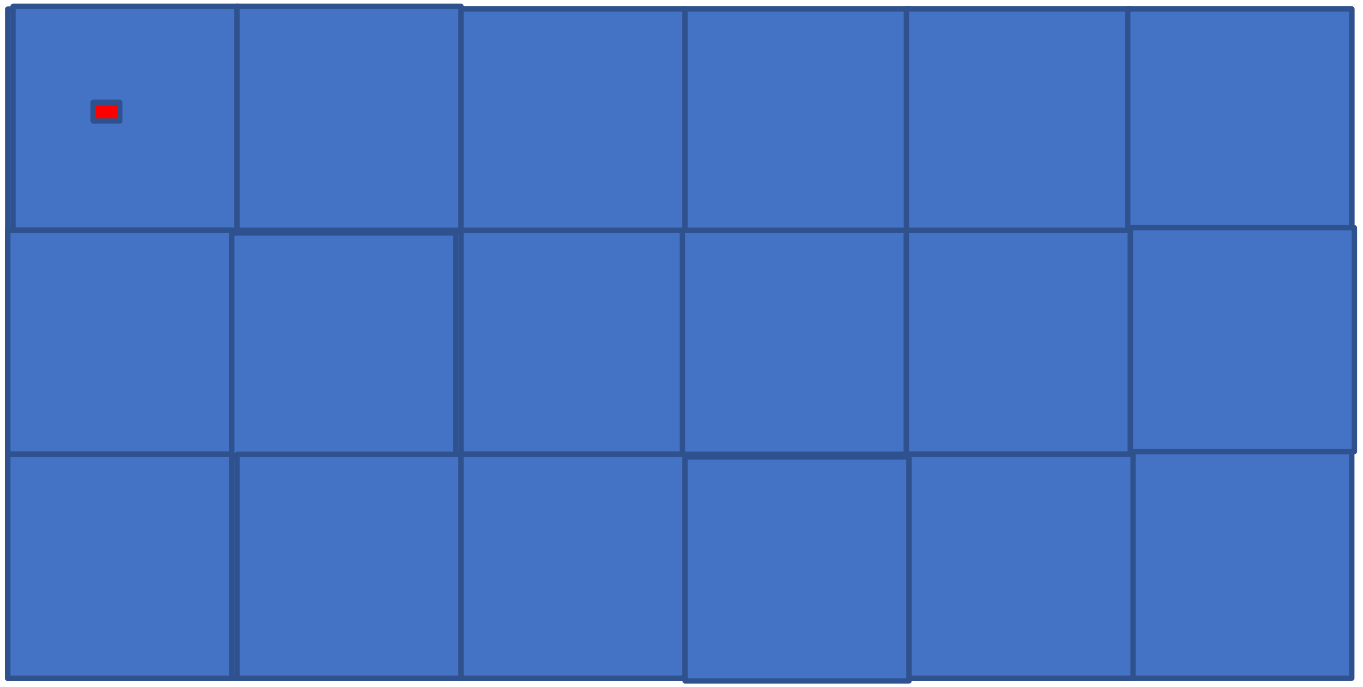
- Load a tile of N into shared memory
  - All threads participate in loading
  - A subset of threads then use each N element in shared memory



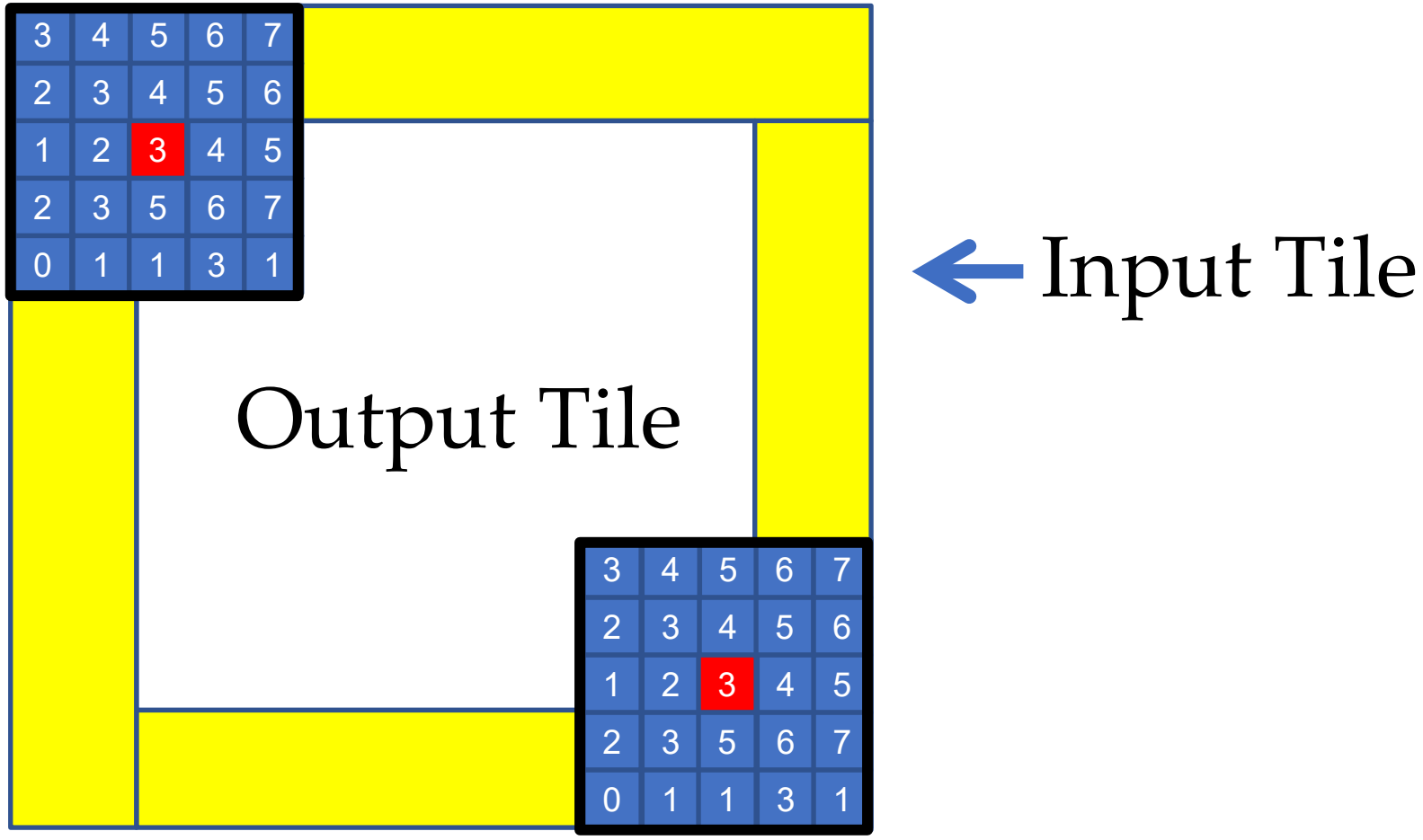
Output Tiles Still Cover the Output!

row_o = blockIdx.y*TILE_WIDTH + threadIdx.y;

col_o = blockIdx.x * TILE_WIDTH + threadIdx.x;



Input Tiles Need to be Larger than Output Tiles



Setting Block Dimensions

```
dim3 dimBlock(TILE_WIDTH + 4,TILE_WIDTH + 4, 1);
```

In general, block width (square blocks) should be

```
     TILE_WIDTH + (MASK_WIDTH-1)
```

```
dim3 dimGrid(ceil(P.width/(1.0*TILE_WIDTH)),
             ceil(P.height/(1.0*TILE_WIDTH)), 1)
```

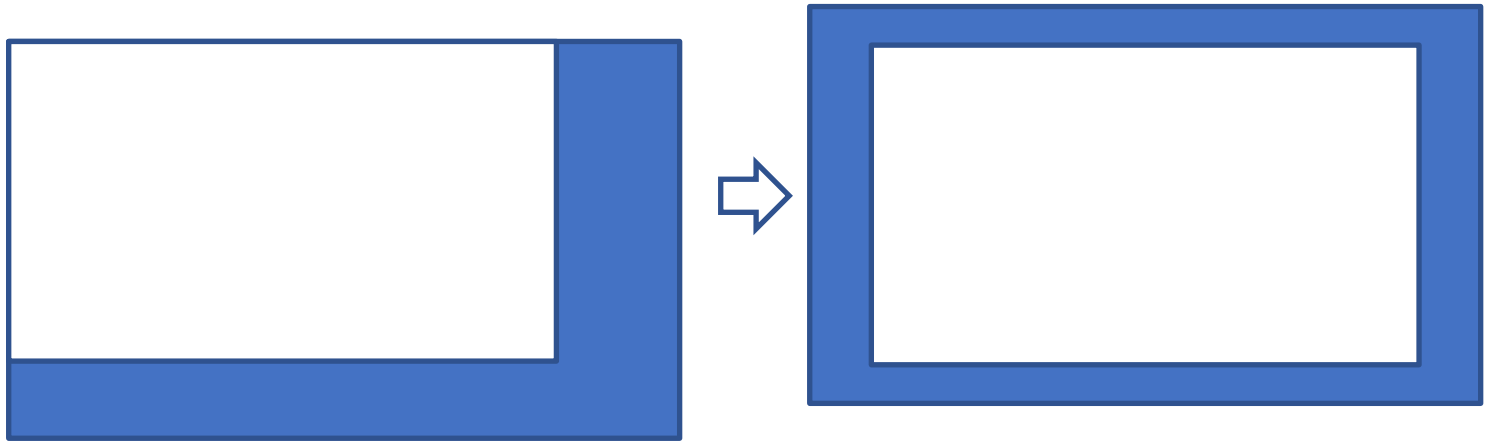
- There need to be enough thread blocks to generate all P elements
- There need to be enough threads to load entire tile of input

Shifting From Output Coordinates To Input Coordinates

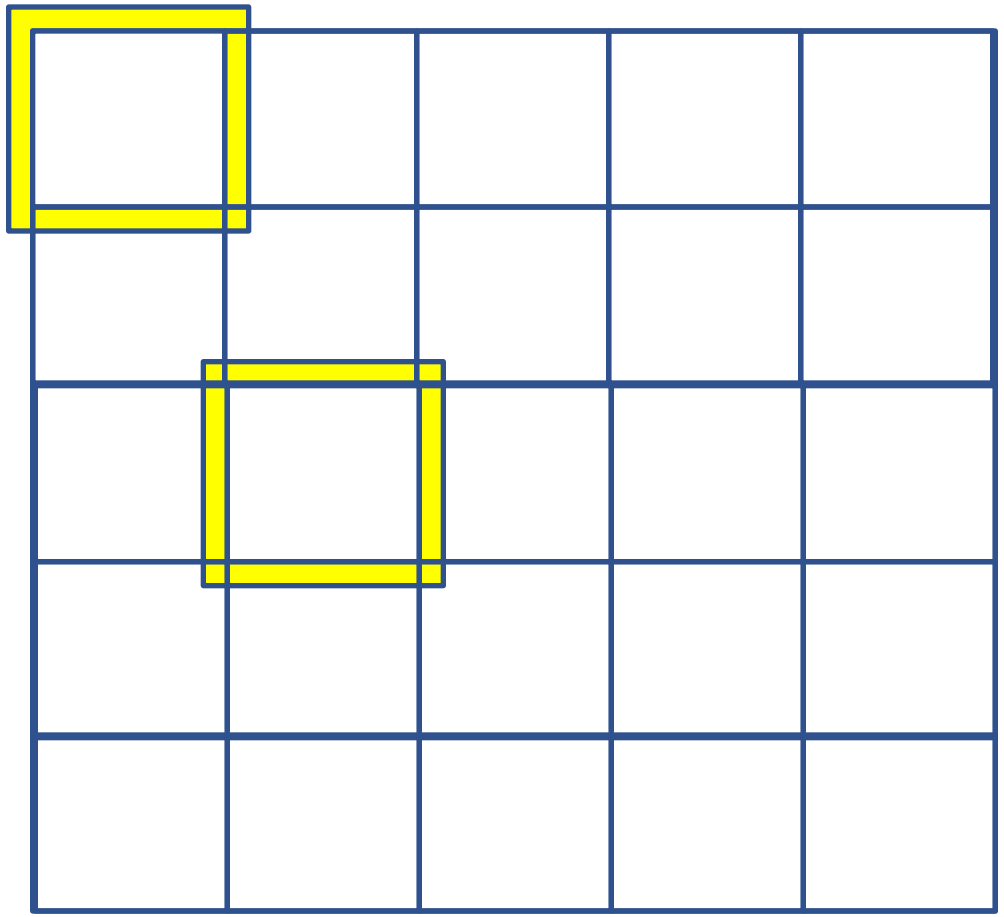
```
int tx = threadIdx.x;
int ty = threadIdx.y;

int row_o = blockIdx.y * TILE_WIDTH + ty;
int col_o = blockIdx.x * TILE_WIDTH + tx;

int row_i = row_o-2; // MASK_WIDTH / 2
int col_i = col_o-2; // (radius in prev. code)
```



Threads That Loads Halos Outside N Should Return 0.0



Taking Care of Boundaries

```
if ((row_i >= 0) && (row_i < Width) &&
    (col_i >= 0)  && (col_i < Width)) {
  tile[ty][tx] = N[row_i*Width + col_i];
} else {
  tile[ty][tx] = 0.0f;
}
__syncthreads (); // wait for tile
```

Not All Threads Calculate and Write Output

```
float Pvalue = 0.0f;
if (ty < TILE_WIDTH && tx < TILE_WIDTH) {
  for(i = 0; i < 5; i++) {
    for(j = 0; j < 5; j++) {
      Pvalue += Mc[i][j] * tile[i+ty][j+tx];
    }
  }
  if (row_o < Width && col_o < Width) {
    P[row_o * Width + col_o] = Pvalue;
  }
}
```

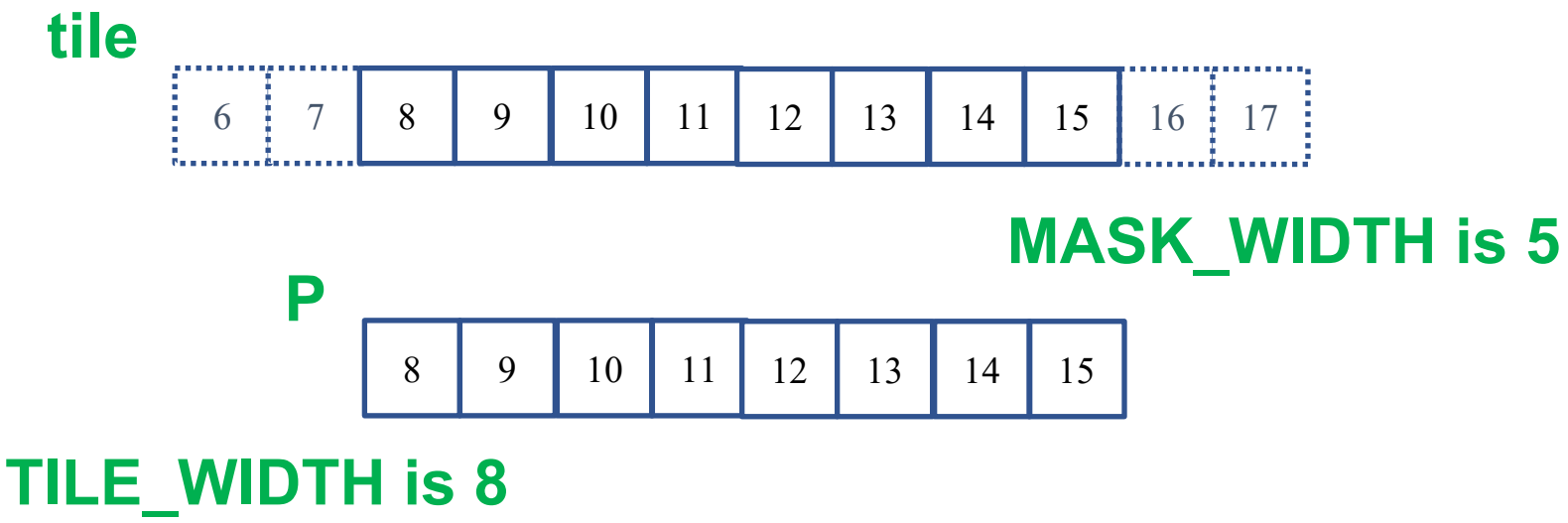
Alternatively

- You can extend the 1D strategy 3 tiled convolution into a 2D strategy 3 tiled convolution
  - Each input tile matches its corresponding output tile
  - All halo elements will be loaded from global memory
  - If condition and divergence during inner product computation

Reuse Analysis

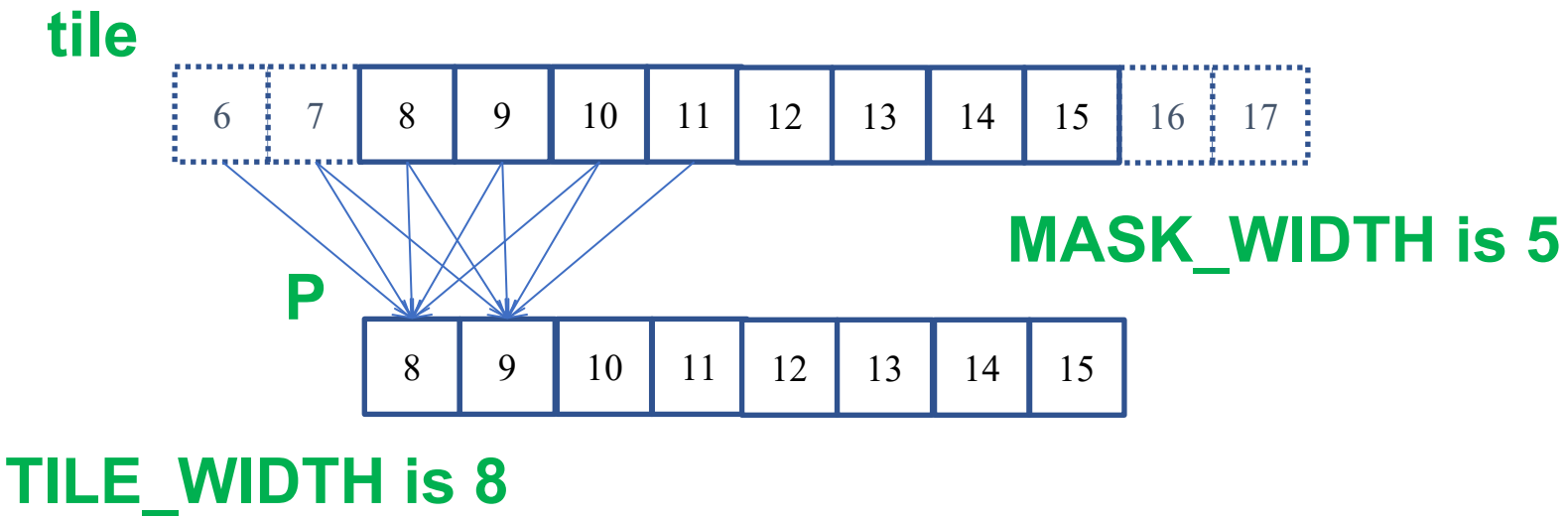
- Let's quantify benefits of tiled convolution
  - Regardless of the kernel implementation strategy

A Small 1D Convolution Example



- output and input tiles for block 1
- For MASK_WIDTH of 5, each block loads
  8 + (5 – 1) = 12 elements (**12 memory loads**)

Each Output Uses MASK_WIDTH Inputs



- P[8] uses N[6], N[7], N[8], N[9], N[10]
- P[9] uses N[7], N[8], N[9], N[10], N[11]
- …
- P[15] uses N[13], N[14], N[15], N[16], N[17]

- Total of **8 * 5 values** from **`tile`** **used for** the **output**

A Simple Way to Calculate Tiling Benefit

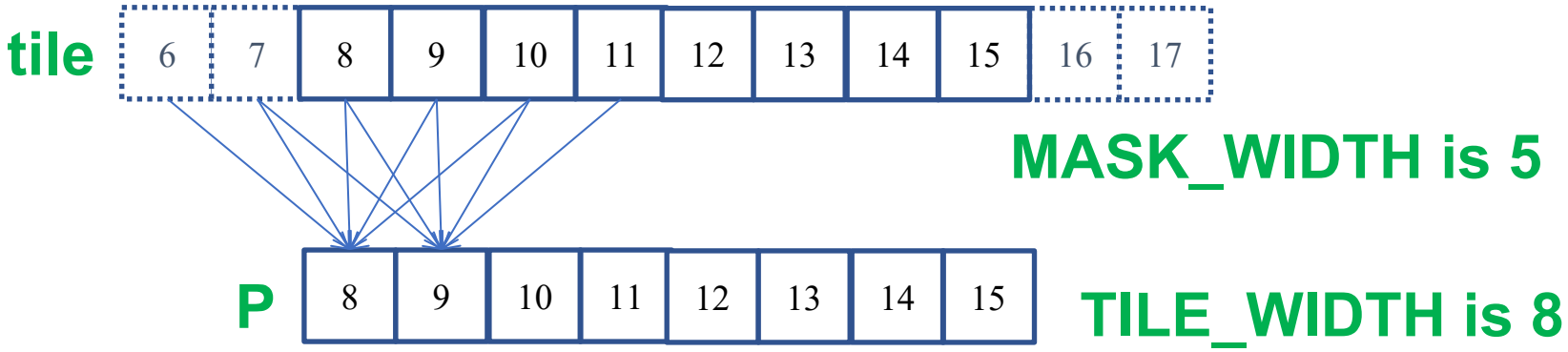
- 8+(5-1) = 12 unique elements of input array N loaded
- 8*5 global memory accesses potentially replaced by shared memory accesses
- This gives a bandwidth reduction of **40/12=3.3**
- This is independent of the size of N

In General, for 1D Convolution Kernels

- Load (TILE_WIDTH + MASK_WIDTH – 1) elements from global memory to shared memory
- Replace (TILE_WIDTH * MASK_WIDTH) global memory accesses with shared memory accesses
- This leads to bandwidth reduction of

(TILE_SIZE * MASK_WIDTH) / (TILE_SIZE + MASK_WIDTH - 1)

Another Way to Look at Reuse



- tile[6] is used by P[8] (1×)
- tile[7] is used by P[8], P[9] (2×)
- tile[8] is used by P[8], P[9], P[10] (3×)
- tile[9] is used by P[8], P[9], P[10], P[11] (4×)
- tile[10] is used by P[8], P[9], P[10], P[11], P[12] (5×)
- … (5×)
- tile[14] is uses by P[12], P[13], P[14], P[15] (4×)
- tile[15] is used by P[13], P[14], P[15] (3×)
- tile[16] is used by P[14], P[15] (2×)
- tile[17] is used by P[15] (1×)

Another Way to Look at Reuse

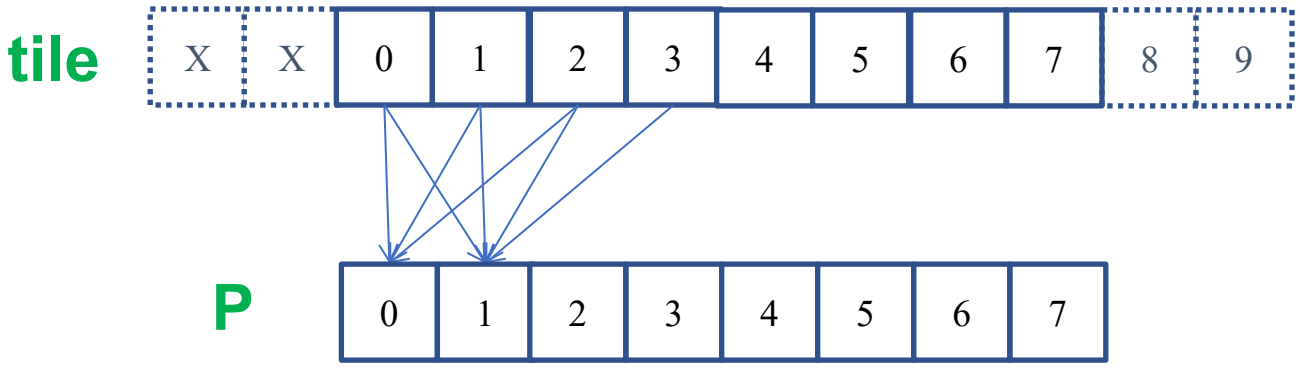
- Each access `tile` replaces an access to input N
- The total number of global memory accesses (to the (8+5-1)=12 input elements) replaced by shared memory accesses is

$$
\begin{aligned}
& 1 + 2 + 3 + 4 + 5 * (8-5+1) + 4 + 3 + 2 + 1 \\
= & 10 + 20 + 10 \\
= & 40
\end{aligned}
$$

- There are 12 elements of input N, so the average reduction is

  40/12 = 3.3

What About Boundary Tiles?



- P[0] uses N[0], N[1], N[2]
- P[1] uses N[0], N[1], N[2], N[3]
- P[2] uses N[0], N[1], N[2], N[3], N[4]

- Less than 8 * 5 elements of N are used for the output tile

Ghost Elements Change Ratios

- For a boundary tile, we load TILE_WIDTH + (MASK_WIDTH-1)/2 elements
  - 10 in our example of TILE_WIDTH of 8 and MASK_WIDTH of 5

- Computing boundary elements do not access global memory for ghost cells
  - Total accesses is 6*5 + 4 + 3 = 37 accesses (when computing the P elements)

- The reduction is **37/10 = 3.7**

In General, for 1D, Internal Tiles

- The total number of global memory accesses to the (TILE_WIDTH+Mask_Width-1) elements of input N replaced by shared memory accesses is

1 + 2 + … + Mask_Width-1+ Mask_Width * (TILE_WIDTH -Mask_Width+1) + Mask_Width-1+… + 2 + 1

= ((Mask_Width-1) *Mask_Width)/2+ Mask_Width*(TILE_WIDTH-Mask_Width+1) + ((Mask_Width-1) *Mask_Width)/2

= (Mask_Width-1)*Mask_Width+ Mask_Width*(TILE_WIDTH-Mask_Width+1)

= Mask_Width*TILE_WIDTH

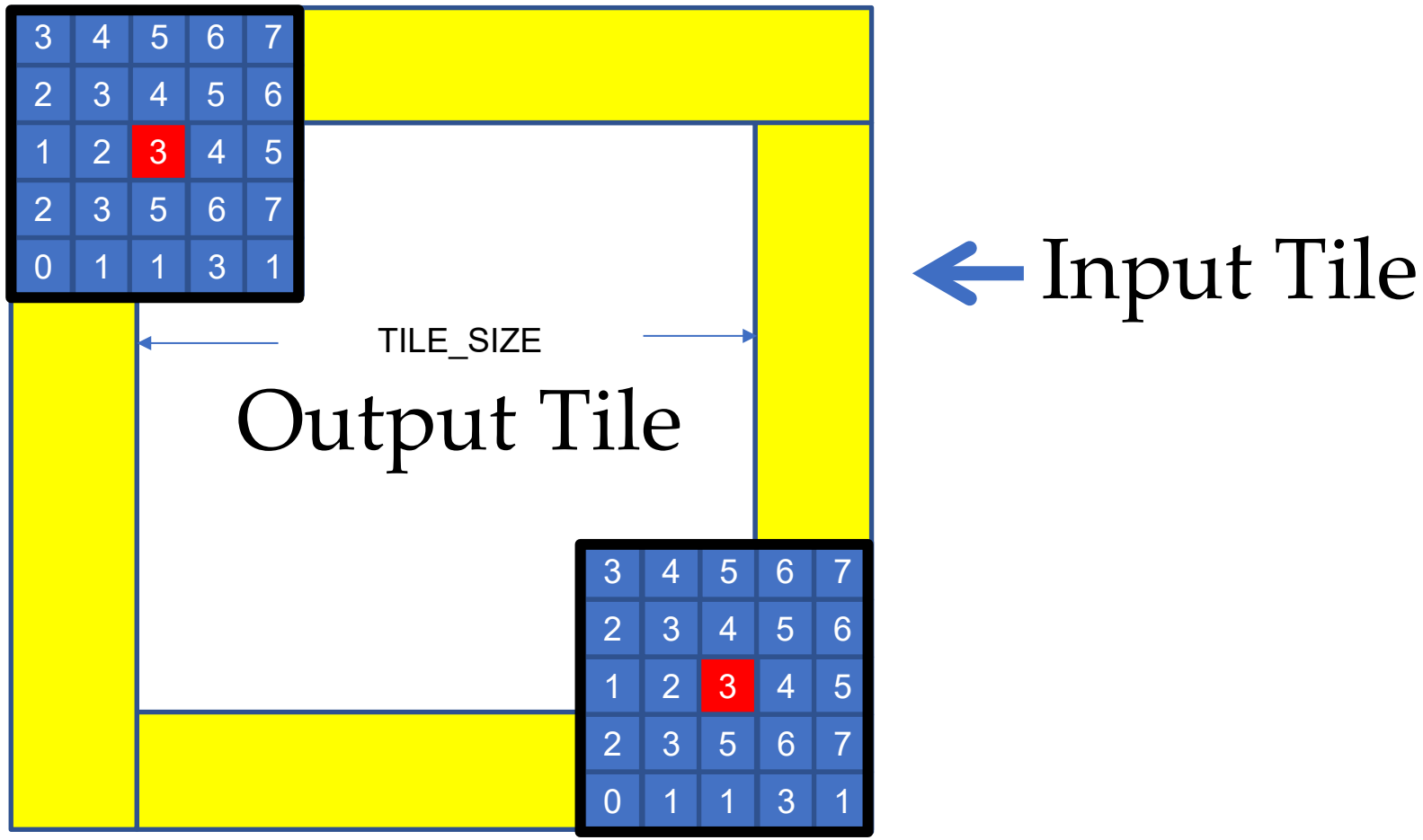
Bandwidth Reduction for 1D

- The reduction is
  - **(TILE_SIZE * MASK_WIDTH) / (TILE_SIZE + MASK_WIDTH - 1)**

TILE_WIDTH	16	32	64	128	256
Reduction MASK_WIDTH = 5	4.0	4.4	4.7	4.9	4.9
Reduction MASK_WIDTH = 9	6.0	7.2	8.0	8.5	8.7

Review: Parallelization of Tile Load

MASK_WIDTH is 5



Analysis for an 8×8 Output Tile, MASK_WIDTH of 5

- Loading input tile requires **$(8+5-1)^2$ = 144 reads**
- Calculation of each output requires 5^2 = 25 input elements
- **8×8×25 = 1,600 global memory accesses** for computing output tile are converted to shared memory accesses
- Bandwidth reduction of **1,600/144 = 11.1×**

In General

- **(TILE_WIDTH+MASK_WIDTH-1)^2 elements** need to be **loaded** from **N** into shared memory
- The calculation of each **P** element needs to access **MASK_WIDTH^2** elements of **N**
- **(TILE_WIDTH * MASK_WIDTH)^2 global memory accesses** converted into shared memory accesses
- Bandwidth reduction of
  (TILE_WIDTH * MASK_WIDTH)^2 / (TILE_WIDTH + MASK_WIDTH - 1)^2

Bandwidth Reduction for 2D Convolution Kernel

- The reduction is
  - **(TILE_WIDTH * MASK_WIDTH)^2 / (TILE_WIDTH + MASK_WIDTH - 1)^2**

TILE_WIDTH	8	16	32	64
Reduction MASK_WIDTH = 5	11.1	16	19.7	22.1
Reduction MASK_WIDTH= 9	20.3	36	51.8	64

Ghost Elements Change Ratios

- 2D calculation left for your enjoyment

2B/FLOP for Untiled Convolution

- **How much global memory per FLOP is in untiled convolution?**

- In untiled convolution,
  - each value from **N** (**4B** from global memory)
  - is multiplied by a value from **M** (**4B** from constant cache, **1 FLOP**),
  - then added to a running sum (**1 FLOP**)

- **That gives 2B / FLOP**

Full Use of Compute Requires 13.3× Reuse

- Recall our reuse discussion from matrix multiply:
  - **1,000 GFLOP/s** for GPU from ~2010, and
  - **150 GB/s** memory bandwidth.

- Dividing memory bandwidth by **2B/FLOP**,

$$\frac{150\ \mathrm{GB/s}}{2\ \mathrm{B/FLOP}} = \mathbf{75\ GFLOP/s = 7.50\%\ of\ peak}.$$

- **Need** at least **100/7.50 = 13.3× reuse** to make full use of compute resources

In 2020, Need 52.1× Reuse

- **In 2020**, the **GRID K520** offers
  - nearly **5,000 GFLOP/s**, but only
  - **192 GB/s** memory bandwidth

- Dividing memory bandwidth by **2B/FLOP**,

$$\frac{192\ \mathrm{GB/s}}{2\ \mathrm{B/FLOP}} = \mathbf{96\ GFLOP/s = 1.92\%\ of\ peak}$$

- **Need** at least **100/1.92 = 52.1× reuse** to make full use of compute resources

Need ~26× Reuse on H100 GPUs

- **In 2023**, the **H100 PCIe** version offers
  - **26 TFLOP/s**, and
  - **2 TB/s** memory bandwidth

- Dividing memory bandwidth by **2B/FLOP**,

$$\frac{2\ \mathrm{TB/s}}{2\ \mathrm{B/FLOP}} = \mathbf{1\ TFLOP/s = 3.85\%\ of\ peak}$$

- **Need** at least **100/3.85 = 25.97× reuse** to make full use of compute resources

Need Really Big Mask to Balance Resources

- Let's make another table: **% of peak compute**
  - for **1D** tiled convolution,
  - with **TILE_WIDTH 1024**

MASK_WIDTH	~2010 GPU with 1,000 GFLOP/s 150 GB/s	~2020 GPU with 5,000 GFLOP/s 192 GB/s
5	37%	9.6%
9	67%	17%
15	100%	28%
55	100%	100%

Need Really Big Mask to Balance Resources

- And one more: **% of peak compute**
  - for **2D** tiled convolution,
  - with **TILE_WIDTH 32×32**

MASK_WIDTH	~2010 GPU with 1,000 GFLOP/s 150 GB/s	~2020 GPU with 5,000 GFLOP/s 192 GB/s
3	60%	15%
5	100%	37%
7	100%	67%
9	100%	almost 100%

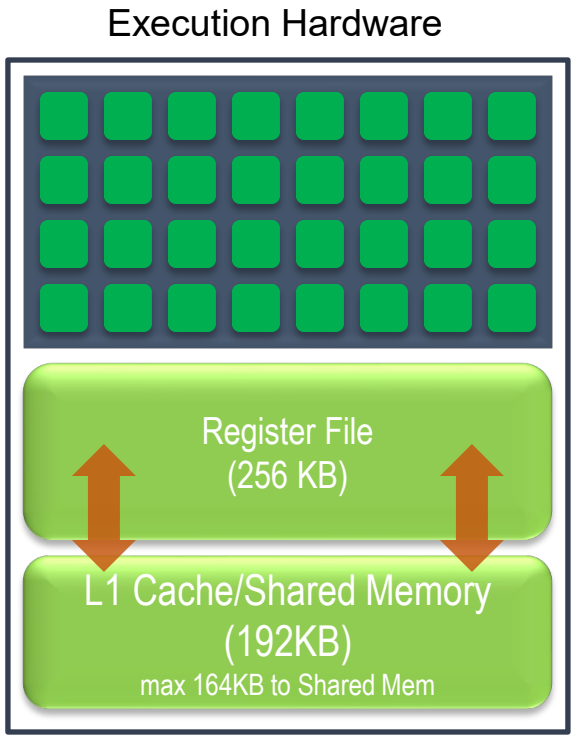
Food for Thought

- Ratios are different for tiles on boundaries
- More importantly,
  - Each thread loads 4B to shared memory
  - 2,048 threads load only 8kB
  - Shared memory is usually 64kB or larger
  - What can one do with the rest?

Improved approach left as homework

(For example, can raise MW=7 from 67% to 81%)

Ampere SM Memory Architecture

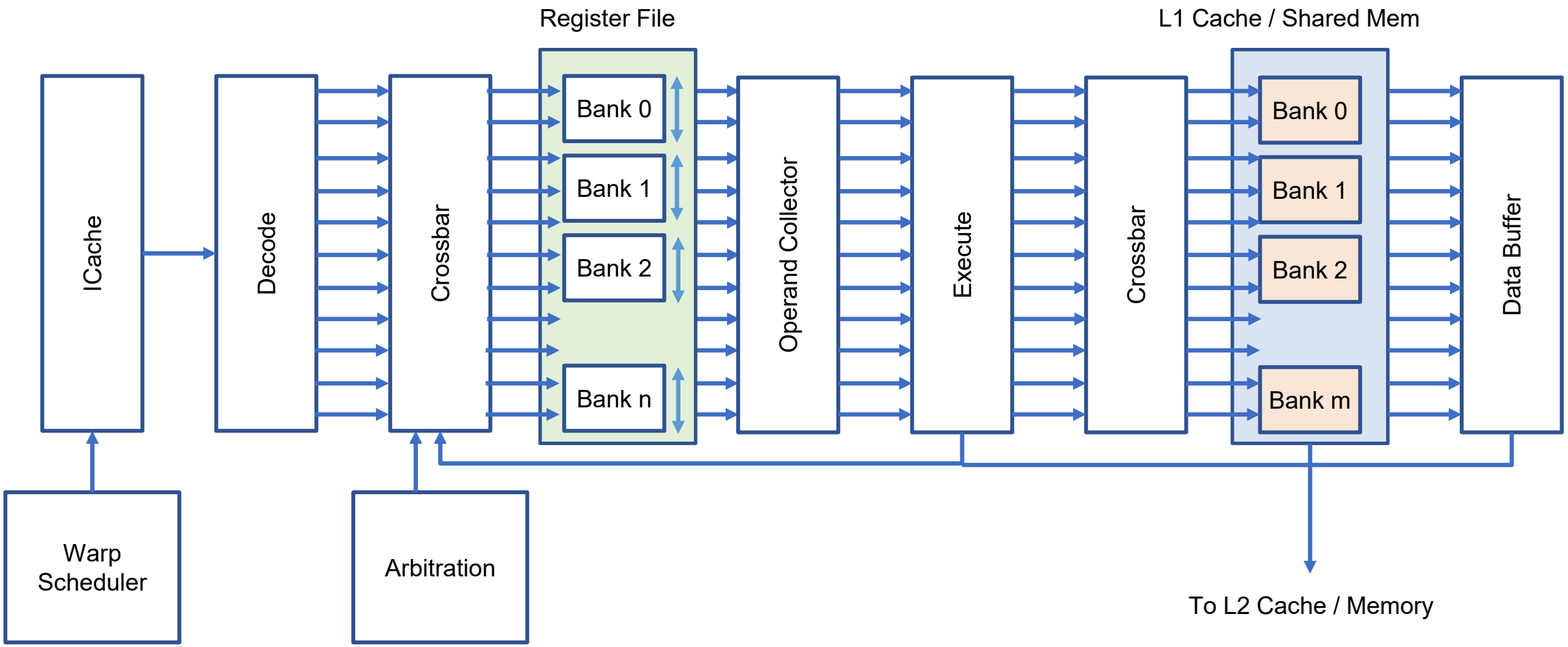


Ampere SM Memories
(GPU from 2020)

- **Memory access time**
  - **registers (~1 cycle)**
  - **shared memory (~5 cycles)**
  - **cache/constant memory (~5 cycles)**
  - **global memory (~500 cycles)**

- **Number of thread blocks mapped to an SM is limited by**
  - **Total number of threads supported by an SM**
  - **Amount of shared memory available on SM**

Overall Data Parallel Pipeline



Memory Hierarchy Considerations

- Register file is highly banked
  - But we can have bank conflicts that cause pipeline stalls
- Shared memory is highly banked
  - But we can have bank conflicts that cause pipeline stalls
- Global memory has multiple channels, banks, pages
  - Relies on bursting
  - Coalescing is important. Need programmer involvement.
- L1 Cache is non-coherent

Any More Questions?

Things to Read:

Textbook chapter 7

CUDA BPG: Memory Optimizations

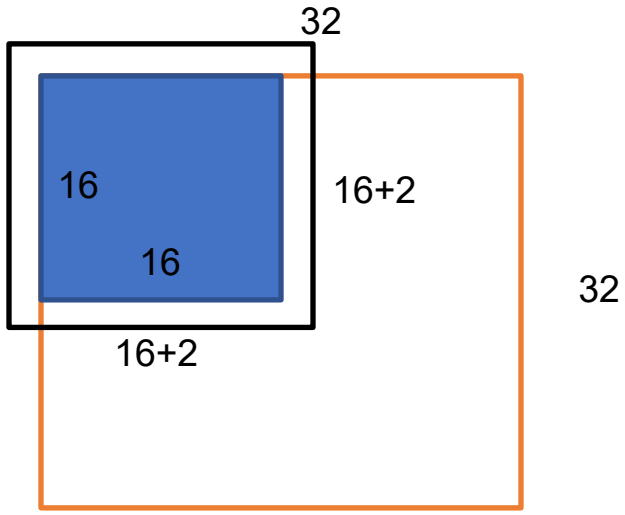
Things to Do:

Submit Lab 4

Sign up for GPT-2 project, if you wish

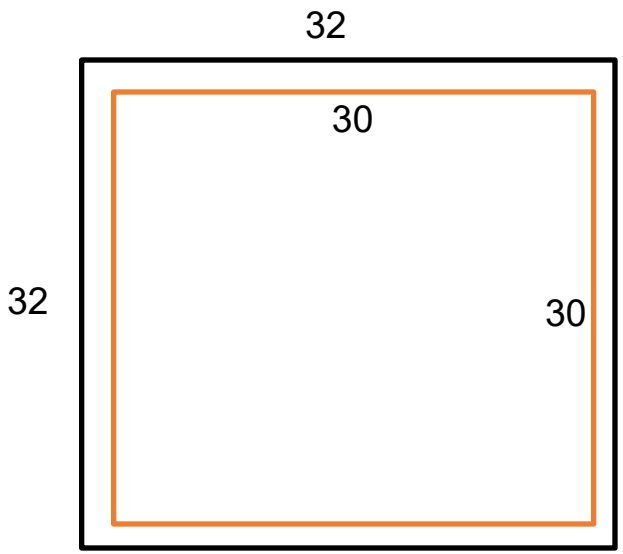
Problem Solving

- Q: Consider 2D tiled convolution with mask width of 5x5 and output tile width of 16x16 applied to an input image of size 32x32. Assume we use Strategy 1. The use of shared memory reduces the number of global memory accesses. What is the reduction in global memory accesses for thread block (0,0)?
- A:
  - Count # of global memory accesses:
    row 0: 9+12+15*14
    row 1: 12+16+20*14
    other 14 rows: 15+20+25*14
  - Count # of input elements to be used in the calculation: 18*18



Problem Solving

- Q: For a tiled 2D convolution kernel with 30x30 output tiles and 3x3 mask (and thus 32x32 input tile), how many warps in each thread block have control divergence? (Assume strategy 2: block size covers input tile.)
- A:
  - thread block size is 32x32
  - All 32x32 threads participate in data load
  - What does each warp do?
    - Warp 0: load 0s and do not compute
    - Warps 1-30: 30 threads compute, and 2 threads do not
    - Warp 31: load 0s and do not compute
  - Thus, only 2 warps do not have control divergence



Problem Solving

- Q: The A40 GPU has a peak FP32 performance of 37.4 TFLOPS and 48 GB of GDDR6 memory with a memory bandwidth of 696 GB/s. These GPUs do not support FP64. What should be the byte-to-FLOP ratio for this GPU to fully utilize its compute capability? How much of data reuse is needed to make full use of the compute resources?
- A:
  - 696 GB/s / 37.4 TFLOPS = 0.019 B/FLOP
  - Another way to look at it: for each byte loaded from global memory, we need to use it in 53.7 floating point operations